PATHOGEN GENOMIC SURVEILLANCE & OUTBREAK-TRACKING SYSTEM

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Pathogen Genomic Surveillance & Outbreak-Tracing System is a data-driven system designed to monitor infectious pathogens, identify potential disease outbreaks, and support the investigation of pathogen transmission. The system combines genomic sequence data with epidemiological information, such as patient location, sample collection date, pathogen type, and case details. Genomic surveillance helps monitor genetic similarities and differences among pathogens, making it easier to identify related cases and possible outbreak clusters.

The proposed system uses bioinformatics techniques to process pathogen genomic data and compare genetic sequences to identify closely related samples. It also organizes cases based on geographical and temporal information, helping users understand how an outbreak may be developing and spreading. Visualization tools such as cluster diagrams, geographical maps, and outbreak timelines can present these findings in a clear and easy-to-understand manner. The system aims to provide an efficient platform for researchers, laboratories, and public-health authorities to monitor pathogen evolution and investigate possible outbreaks. Whole-genome sequencing provides detailed genetic information that can improve outbreak detection and investigation compared with older, lower-resolution methods.

By combining genomic analysis, epidemiological data, and visualization, the system can help identify outbreak clusters more quickly and provide publichealth teams with useful information for making informed decisions related to disease-control activities.

Introduction

Infectious diseases can spread rapidly, making it difficult to identify related cases, find possible sources of an outbreak, and understand how the disease is spreading. Traditional disease surveillance mainly depends on clinical and epidemiological information such as patient details, location, and sample collection dates. However, genomic data can provide additional information by showing how closely different pathogen samples are related.
The Pathogen Genomic Surveillance & Outbreak-Tracing System combines pathogen genomic sequences with patient, location, and time-based information
to identify genetically related cases and possible outbreak clusters. By comparing sequences from different samples, the system can identify similarities and differences and help understand the spread of a pathogen. The project mainly applies String Algorithms and Dynamic Programming concepts from DSA-3. Needleman-Wunsch and Smith-Waterman algorithms are used for global and local sequence alignment, while the Wagner-Fischer algorithm is used to measure differences between sequences. These algorithms help determine the similarity between pathogen samples and support the identification of related cases.

The system presents the analyzed information through outbreak clusters, geographical maps, and timelines. By combining genomic analysis with epidemiological data, the system can help users understand pathogen transmission, monitor possible outbreaks, and support outbreak investigation.

Software and hardware requirements

Category	Requirement
OS	Windows 10/11 or Ubuntu
Processor	Intel i5 / Ryzen 5 or higher
RAM	8 GB minimum
Storage	256 GB SSD or higher
Programming	Python 3.x
Libraries	Pandas, NumPy, Biopython, Scikit-learn
Frontend	HTML, CSS, JavaScript
Backend	Flask / FastAPI
Database	MySQL / PostgreSQL
Tools	Jupyter / VS Code, Git
Internet	Stable Internet connection

Signature of the Guide   Course Instructor